PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints.(**Note**- Create Table, insert data as per your Choice)

```
mysql> create table Department(dept_id int primary key auto_increment,dept_name varchar(50),city varchar(50));
Query OK, 0 rows affected (0.04 sec)

mysql> desc department;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| dept_id   | int         | NO   | PRI | NULL    | auto_increment |
| dept_name | varchar(50) | YES  |     | NULL    |                |
| city      | varchar(50) | YES  |     | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
3 rows in set (0.00 sec)
```

```
mysql> create table Employee(emp_id int primary key auto_increment,name varchar(20),salary int,dept_id int,
    -> constraint fk_dept_id foreign key (dept_id) references Department(dept_id) on delete cascade);
Query OK, 0 rows affected (0.06 sec)

mysql> desc Employee;
+---------+-------------+------+-----+---------+----------------+
| Field   | Type        | Null | Key | Default | Extra          |
+---------+-------------+------+-----+---------+----------------+
| emp_id  | int         | NO   | PRI | NULL    | auto_increment |
| name    | varchar(20) | YES  |     | NULL    |                |
| salary  | int         | YES  |     | NULL    |                |
| dept_id | int         | YES  | MUL | NULL    |                |
+---------+-------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

2. Write a SQL query to add any new column to an existing table.

```
mysql> alter table Employee
    -> add joining_date date;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employee;
+--------------+-------------+------+-----+---------+----------------+
| Field        | Type        | Null | Key | Default | Extra          |
+--------------+-------------+------+-----+---------+----------------+
| emp_id       | int         | NO   | PRI | NULL    | auto_increment |
| name         | varchar(20) | YES  |     | NULL    |                |
| salary       | int         | YES  |     | NULL    |                |
| dept_id      | int         | YES  | MUL | NULL    |                |
| joining_date | date        | YES  |     | NULL    |                |
+--------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

```
mysql> insert into Department(dept_name,city) values ('Technical','Mumbai'),('Hr','Nagpur'),('Finance','Mumbai'),('Techn
ical','Nagpur'),('R&D','Banglore');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from department;
+---------+-----------+----------+
| dept_id | dept_name | city     |
+---------+-----------+----------+
|       1 | Technical | Mumbai   |
|       2 | Hr        | Nagpur   |
|       3 | Finance   | Mumbai   |
|       4 | Technical | Nagpur   |
|       5 | R&D       | Banglore |
+---------+-----------+----------+
5 rows in set (0.00 sec)
```

```
mysql> insert into Employee (name,salary,dept_id)
    -> values ('Rohit',50000,1),
    -> ('Nikhil',40000,2),
    -> ('Suraj',45000,3),
    -> ('Amol',30000,3),
    -> ('Pranit',43000,5);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Employee;
+--------+--------+--------+---------+
| emp_id | name   | salary | dept_id |
+--------+--------+--------+---------+
|      1 | Rohit  |  50000 |       1 |
|      2 | Nikhil |  40000 |       2 |
|      3 | Suraj  |  45000 |       3 |
|      4 | Amol   |  30000 |       3 |
|      5 | Pranit |  43000 |       5 |
+--------+--------+--------+---------+
5 rows in set (0.00 sec)
```

4. Write a SQL query to delete all records from a table.

```
mysql> delete from employee;
Query OK, 7 rows affected (0.00 sec)

mysql> select * from employee;
Empty set (0.00 sec)
```

5. Write a SQL query to update salary records from a table where id = 5.

```
mysql> update Employee set salary = 35000 where emp_id = 5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Employee;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      1 | Rohit  |  50000 |       1 | NULL         |
|      2 | Nikhil |  40000 |       2 | NULL         |
|      3 | Suraj  |  45000 |       3 | NULL         |
|      4 | Amol   |  30000 |       3 | NULL         |
|      5 | Pranit |  35000 |       5 | NULL         |
|      6 | Mohit  |  60000 |       4 | NULL         |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.

```
mysql> create user 'readonly_user'@'localhost'
    -> identified by 'Pass@1234';
Query OK, 0 rows affected (0.02 sec)

mysql> grant select on Employee
    -> to 'readonly_user'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> show grants for 'readonly_user'@'localhost';
+-----------------------------------------------------------------+
| Grants for readonly_user@localhost                              |
+-----------------------------------------------------------------+
| GRANT USAGE ON *.* TO `readonly_user`@`localhost`               |
| GRANT SELECT ON `cdac`.`employee` TO `readonly_user`@`localhost` |
+-----------------------------------------------------------------+
2 rows in set (0.00 sec)
```

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.

```
mysql> create user 'temp_user'@'localhost'
    -> identified by 'Pass@1234';
Query OK, 0 rows affected (0.02 sec)

mysql> grant insert on Employee to
    -> 'temp_user'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> show grants for 'temp_user'@'localhost';
+-----------------------------------------------------------+
| Grants for temp_user@localhost                            |
+-----------------------------------------------------------+
| GRANT USAGE ON *.* TO `temp_user`@`localhost`             |
| GRANT INSERT ON `cdac`.`employee` TO `temp_user`@`localhost` |
+-----------------------------------------------------------+
2 rows in set (0.00 sec)

mysql> revoke insert from 'temp_user'@'localhost';
ERROR 1064 (42000): Illegal authorization identifier near 'insert from 'temp_user'@'localhost'' at line 1
mysql> revoke insert on Employee from 'temp_user'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> show grants for 'temp_user'@'localhost';
+-------------------------------------------+
| Grants for temp_user@localhost            |
+-------------------------------------------+
| GRANT USAGE ON *.* TO `temp_user`@`localhost` |
+-------------------------------------------+
1 row in set (0.00 sec)
```

8. Write a query to insert a new employee into the Emp table and commit the transaction.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into Employee(name,salary,dept_id) values ('Mohit',60000,4);
Query OK, 1 row affected (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from Employee;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      1 | Rohit  |  50000 |       1 | NULL         |
|      2 | Nikhil |  40000 |       2 | NULL         |
|      3 | Suraj  |  45000 |       3 | NULL         |
|      4 | Amol   |  30000 |       3 | NULL         |
|      5 | Pranit |  43000 |       5 | NULL         |
|      6 | Mohit  |  60000 |       4 | NULL         |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

9. Write a query to update the Salary of an employee with EmpID = 101 and commit the transaction.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update Employee set salary = 58000 where emp_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from Employee;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      1 | Rohit  |  58000 |       1 | NULL         |
|      2 | Nikhil |  40000 |       2 | NULL         |
|      3 | Suraj  |  45000 |       3 | NULL         |
|      4 | Amol   |  30000 |       3 | NULL         |
|      5 | Pranit |  35000 |       5 | NULL         |
|      6 | Mohit  |  60000 |       4 | NULL         |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> savepoint sp1;
Query OK, 0 rows affected (0.00 sec)

mysql> update Employee set dept_id = 1 where emp_id = 5;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Employee;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      1 | Rohit  |  58000 |       1 | NULL         |
|      2 | Nikhil |  40000 |       2 | NULL         |
|      3 | Suraj  |  45000 |       3 | NULL         |
|      4 | Amol   |  30000 |       3 | NULL         |
|      5 | Pranit |  35000 |       1 | NULL         |
|      6 | Mohit  |  60000 |       4 | NULL         |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

11. Write a query to rollback to the previously created SAVEPOINT.

```
mysql> rollback to sp1;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from Employee;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      1 | Rohit  |  58000 |       1 | NULL         |
|      2 | Nikhil |  40000 |       2 | NULL         |
|      3 | Suraj  |  45000 |       3 | NULL         |
|      4 | Amol   |  30000 |       3 | NULL         |
|      5 | Pranit |  35000 |       5 | NULL         |
|      6 | Mohit  |  60000 |       4 | NULL         |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

12. Write a query to truncate all data from the Emp table.

```
mysql> truncate employee;
Query OK, 0 rows affected (0.08 sec)
```

13. Write a query to list all employees whose salary is greater than 50,000.

```
mysql> select * from Employee where salary > 50000;
+--------+-------+--------+---------+--------------+
| emp_id | name  | salary | dept_id | joining_date |
+--------+-------+--------+---------+--------------+
|      1 | Rohit |  58000 |       1 | NULL         |
|      6 | Mohit |  60000 |       4 | NULL         |
+--------+-------+--------+---------+--------------+
2 rows in set (0.00 sec)
```

14. Write a query to increase the salary of all employees in DeptID = 10 by 10%.

```
mysql> update Employee set salary = salary * 1.10 where emp_id = 4;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Employee;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      1 | Rohit  |  58000 |       1 | NULL         |
|      2 | Nikhil |  40000 |       2 | NULL         |
|      3 | Suraj  |  45000 |       3 | NULL         |
|      4 | Amol   |  33000 |       3 | NULL         |
|      5 | Pranit |  35000 |       5 | NULL         |
|      6 | Mohit  |  60000 |       4 | NULL         |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

15. Write a query to count the total number of employees in each department.

```
mysql> select dept_id,count(*) as total_employee
    -> from employee
    -> group by dept_id;
+---------+----------------+
| dept_id | total_employee |
+---------+----------------+
|    NULL |              1 |
|       1 |              1 |
|       2 |              1 |
|       3 |              2 |
|       4 |              1 |
|       5 |              1 |
+---------+----------------+
6 rows in set (0.00 sec)
```

16. Write a query to list departments that have no employees. & create a view for those salespeople who belong to the city of Mumbai.

```
mysql> select * from department d
    -> left join Employee e
    -> on d.dept_id = e.dept_id
    -> where e.emp_id is null;
+---------+----------------+--------+--------+------+--------+---------+--------------+
| dept_id | DepartmentName | city   | emp_id | name | salary | dept_id | joining_date |
+---------+----------------+--------+--------+------+--------+---------+--------------+
|       6 | Sales          | Mumbai |   NULL | NULL |   NULL |    NULL | NULL         |
+---------+----------------+--------+--------+------+--------+---------+--------------+
1 row in set (0.00 sec)
```

```
mysql> select e.* from employee e
    -> join department d
    -> on e.dept_id = d.dept_id
    -> where d.departmentName = 'Technical' and
    -> d.city = 'Mumbai';
+--------+-------+--------+---------+--------------+
| emp_id | name  | salary | dept_id | joining_date |
+--------+-------+--------+---------+--------------+
|      1 | Rohit |  58000 |       1 | 2002-11-19   |
+--------+-------+--------+---------+--------------+
1 row in set (0.00 sec)
```

17. Write a query to display the highest salary among all employees.

```
mysql> select max(salary) as highest_salary from employee;
+----------------+
| highest_salary |
+----------------+
|          60000 |
+----------------+
1 row in set (0.00 sec)
```

```
mysql> select emp_id,name,salary as Highest_salary from employee order by salary desc limit 1 ;
+--------+-------+----------------+
| emp_id | name  | Highest_salary |
+--------+-------+----------------+
|      6 | Mohit |          60000 |
+--------+-------+----------------+
1 row in set (0.00 sec)
```

18. Write a query to display the 2nd highest salary among all employees.

```
mysql> select max(salary) from employee
    -> where salary < (
    -> select max(salary) from employee);
+-------------+
| max(salary) |
+-------------+
|       58000 |
+-------------+
1 row in set (0.01 sec)
```

```
mysql> select emp_id,name,salary as 2nd_Highest_salary from employee order by salary desc limit 1 offset 1;
+--------+-------+--------------------+
| emp_id | name  | 2nd_Highest_salary |
+--------+-------+--------------------+
|      1 | Rohit |              58000 |
+--------+-------+--------------------+
1 row in set (0.00 sec)
```

19. Write a query to show all employees ordered by JoiningDate in descending order.

```
mysql> select * from employee order by joining_date desc;
+--------+--------+--------+---------+--------------+
| emp_id | name   | salary | dept_id | joining_date |
+--------+--------+--------+---------+--------------+
|      6 | Mohit  |  60000 |       4 | 2008-04-22   |
|      2 | Nikhil |  40000 |       2 | 2005-10-19   |
|      1 | Rohit  |  58000 |       1 | 2002-11-19   |
|      5 | Pranit |  35000 |       5 | 1997-10-10   |
|      4 | Amol   |  33000 |       3 | 1993-09-17   |
|      3 | Suraj  |  45000 |       3 | 1993-07-09   |
+--------+--------+--------+---------+--------------+
6 rows in set (0.00 sec)
```

20. Write a query to rename the DeptName column to DepartmentName.

```
mysql> alter table Department
    -> rename column dept_name to DepartmentName;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from department;
+---------+----------------+-----------+
| dept_id | DepartmentName | city      |
+---------+----------------+-----------+
|       1 | Technical      | Mumbai    |
|       2 | Hr             | Nagpur    |
|       3 | Finance        | Mumbai    |
|       4 | Technical      | Nagpur    |
|       5 | R&D            | Banglore  |
+---------+----------------+-----------+
5 rows in set (0.00 sec)
```